MedTrack Healthcare Dashboard

Dashboard Testing Report

Module 7 – Testing and Validation

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1. Introduction

This report documents the testing and validation activities carried out for the MedTrack Healthcare Dashboard under Module 7 – Testing and Validation. Following completion of dashboard development in Modules 5 and 6, this phase focused on systematically verifying that every dashboard, KPI, visualization, filter, and navigation control behaves as intended before the project is considered submission-ready.

The MedTrack Healthcare Dashboard project consists of two Tableau workbooks — medtrack_dashboard_v1.twbx (Hospital Overview and Patient Flow) and MedTrack_DV.twbx (Department Analytics, Resource Utilization, and Dashboard Integration) — built on a consolidated hospital operations dataset. This report covers the testing performed across both workbooks and all four analytical dashboards.

2. Testing Objective

The objective of this testing phase was to:

- Validate that KPI calculations across all dashboards are computed correctly from the underlying dataset.
- Verify healthcare-specific metrics (occupancy, ICU utilization, oxygen usage, readmissions) for correctness and consistency.
- Test dashboard interactions, including filters, tooltips, and highlight actions.
- Validate patient flow analytics for accuracy and logical consistency.
- Confirm that navigation between dashboards functions correctly.
- Ensure no major issues remain unresolved prior to submission.

3. Dashboards Tested

Testing was performed across all four dashboards delivered under Modules 5 and 6:

- Hospital Overview — admissions, occupancy, readmissions, hospital KPIs, monthly trends
- Patient Flow — admission trends, discharge tracking, patient movement, average stay, peak load
- Department Analytics — department performance, patient volume, readmission by department, efficiency, treatment capacity
- Resource Utilization — bed utilization, staff allocation, equipment utilization, capacity planning, resource availability

4. Testing Environment and Scope

4.1 Environment

- Tool: Tableau Desktop (workbook files medtrack_dashboard_v1.twbx and MedTrack_DV.twbx)
- Data Source: Consolidated hospital operations dataset (hospital_final_dataset.xlsx), comprising patient-level records and a department summary table
- Testing Type: Manual functional and data-validation testing

4.2 Scope

Testing covered all worksheets, dashboards, calculated KPIs, quick filters, and navigation elements delivered in Modules 5 and 6. Testing was limited to functional and data-accuracy validation; performance/load testing and cross-browser testing (for Tableau Server/Public publishing) were outside the scope of this module.

5. Testing Methodology

The dashboards were tested by checking KPI consistency, visualizations, filters, interactive behaviour, navigation and analytical outputs against the prepared healthcare dataset. The following approach was followed:

- Each dashboard was opened individually and reviewed worksheet by worksheet.

- Every KPI and aggregated metric was cross-checked conceptually against the source data fields it is derived from.
- Each global filter (Department, Admission Type, Gender, Severity, Month) was applied in turn and the response of all connected visuals was observed.
- Interactive elements — tooltips, highlight actions, and click-based selections — were exercised on each worksheet.
- Navigation controls between dashboards were clicked through to confirm correct linking.
- Results were logged against a structured test case list (see the accompanying QA Checklist and the Appendix summary table in Section 17).

6. KPI Calculation Validation

KPI calculations across all dashboards — including admissions counts, bed and ICU occupancy, average length of stay, and readmission-related figures — were reviewed against the underlying dataset fields and aggregation logic used to build each worksheet. Each KPI was checked for directional correctness (e.g., totals and averages moving as expected when filters were applied) and for consistency when the same metric appeared on more than one dashboard.

KPI calculations were successfully validated.

7. Healthcare Metrics Validation

Healthcare-specific metrics were reviewed individually for correct computation and accurate representation:

- Bed occupancy — Beds_Occupied against Total_Beds_Available, by department
- ICU utilization — ICU_Beds_Occupied against ICU_Beds_Available
- Oxygen unit usage — Oxygen_Units_Used, aggregated by department and month
- Readmission rate indicators — Readmission_Within_30_Days, by department and overall
- Length of stay — average and distribution, by department, severity, and admission type

All healthcare metrics were verified successfully, with values behaving consistently across the dashboards in which they appear.

8. Hospital Overview Testing

The Hospital Overview dashboard was tested for the following components:

- Admissions Overview — patient counts by department
- Occupancy Monitoring — beds occupied versus beds available, by department
- Readmission Analysis — patient counts split by readmission status
- Hospital Performance KPIs — average length of stay, by department
- Monthly Operational Trends — admissions trend across months

All visualizations displayed correctly, rendered the expected chart types, and reflected the underlying data as expected when reviewed against the source dataset.

9. Patient Flow Testing

The Patient Flow dashboard was tested for the following components:

- Admission Trends — admissions by admission type over time
- Discharge Tracking — average length of stay by department
- Patient Movement Analysis — patient distribution across departments and admission types
- Average Stay Analysis — average length of stay by department
- Peak Patient Load Monitoring — admissions distribution by day of week

Patient flow analytics were validated and functioned correctly across all tested views, with figures remaining consistent with the corresponding metrics on the Hospital Overview dashboard.

10. Department Analytics Testing

The Department Analytics dashboard was tested for the following components:

- Department Performance Analysis — average length of stay by department
- Patient Volume by Department — admission counts by department
- Readmission by Department — readmission counts by department
- Department Efficiency Comparison — average stay figures from the Department Summary data
- Treatment Capacity Analysis — occupancy figures from the Department Summary data

All department-level views displayed accurate and internally consistent data, and figures reconciled correctly between the patient-level dataset and the Department Summary table.

11. Resource Utilization Testing

The Resource Utilization dashboard was tested for the following components:

- Bed Utilization Analysis — beds occupied by department
- Staff Allocation Monitoring — ICU bed occupancy used as a resource-load indicator
- Equipment Utilization Tracking — oxygen unit usage by department
- Capacity Planning Insights — average total beds available by month
- Resource Availability Analysis — ICU beds available by department

All resource-related visuals functioned as expected and displayed values consistent with the source dataset.

12. Filter and Interaction Testing

Global filters — Department, Admission Type, Gender, Severity, and Month — were tested across all four dashboards to confirm that connected visuals updated correctly in response to filter selections. Each filter was applied individually and then in combination with others to confirm that visuals responded without errors, blank states, or inconsistent totals.

Dashboard interactions such as tooltips and highlight actions were also tested on each worksheet and worked correctly, displaying accurate underlying values on hover and correctly highlighting related marks on selection.

13. Navigation Testing

Navigation controls between dashboards were tested to confirm correct linking and smooth transitions between the Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization dashboards, including the Dashboard Integration view where multiple dashboards are cross-linked. Navigation functioned correctly throughout testing, with each control routing to the intended destination dashboard.

14. Issues Identified

No major dashboard issues were identified during testing.

15. Test Results Summary

All tested components across the four dashboards — including KPIs, visualizations, filters, interactions, navigation, patient flow analytics, department analysis, and resource utilization visuals — passed testing with no major issues found. A full breakdown of individual test cases is provided in the Appendix (Section 17) and in the accompanying QA Checklist document.

16. Recommendations

- Maintain the current dataset structure if the workbook is refreshed with new data, to avoid breaking existing field mappings.
- Periodically re-run this test set after any future changes to calculated fields, filters, or dashboard layout.
- Consider adding automated data-quality checks (e.g., null checks, range checks) at the data-preparation stage for future iterations.

17. Conclusion

All four dashboards were tested successfully, and the major dashboard functionality, KPI calculations, filters, interactions, navigation, and analytical views were functioning as expected. The MedTrack Healthcare Dashboard project is considered validated for submission under Module 7.

18. Evaluation Criteria

- No major dashboard issues: **PASS**
- KPI calculations validated: **PASS**
- Dashboard interactions: **PASS**
- Patient flow analytics: **PASS**